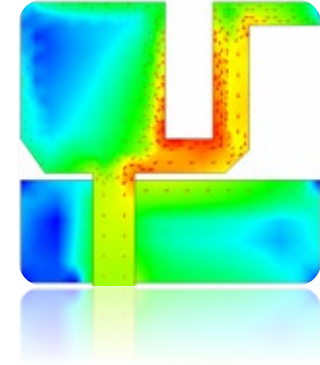




Antenna Engineering

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Assignment II

1. All solutions must be in A4 paper in PDF
2. Submission to folder for Assignment II thru Canvas (only once)
3. Submission deadline is 22 November 2024
4. File name: AII-(Student ID)-(Student Name)

Assignment II-Q1

1. Consider four short dipoles of length ΔL , carrying triangular currents $\mathbf{I}=\mathbf{a}_z I$ with maximum amplitude of I_0 . The dipoles are positioned along z-direction as shown in figure and with radiation pattern of $F(\theta)=\sin \theta$. If phase shift and spacing between adjacent elements are, respectively $\beta=0$ and (1) $d=\lambda/2$ and (2) $d=3\lambda/4$, in the x-z plane
- find electric fields in the far-field zone;
 - sketch radiation patterns;
 - calculate the main beam directions;
 - calculate the null directions;
 - set phase shift β for main beam direction at $\theta=45^\circ$; and
 - check if there is any grating lobe as $\beta \neq 0$. If yes, set the range of phaseshift β to avoid the grating lobes in the visible regions of TWO arrays.

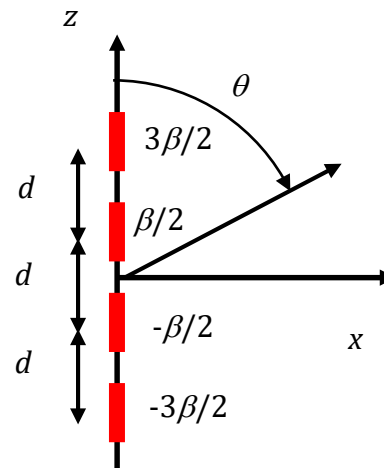


Figure for Problem 1

Assignment II-Q2

2. A two-dimensional, uniformly excited array of isotropic elements as shown below is to be analyzed. Use the principle of pattern multiplication with a pair of elements oriented vertically as an “array” and the four elements in a row as an “element.” Plot the patterns in the xz , yz and xy -planes.

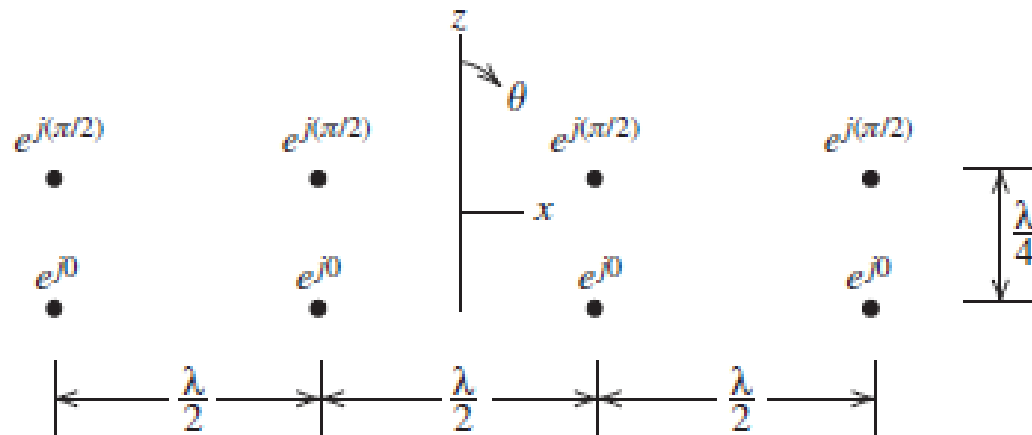


Figure for Problem 2